AI-Powered Restaurant Operations & Review Assistant

Overview

Restaurant businesses regularly handle recurring customer issues involving food quality, service experience, waiting time, cleanliness, pricing, staff behavior, and overall customer satisfaction. Over time, restaurant organizations accumulate thousands of customer reviews, restaurant records, ratings, and business attributes containing valuable information about customer experiences and operational issues.

However, manually searching through historical customer reviews to identify recurring complaints, understand their causes, and determine appropriate corrective actions can be time-consuming and inefficient.

Restaurant managers often ask questions such as:

• Customers are repeatedly complaining about slow service at a restaurant. What similar complaints have been reported before, and what issues are associated with them?

• Multiple restaurants are receiving low ratings despite having similar customer profiles. What operational factors could be contributing to the poor ratings?

• A restaurant is receiving an increasing number of complaints about food quality and staff service. Which issues should be prioritized for corrective action?

Instead of manually searching through historical customer reviews and restaurant records, restaurant managers should be able to quickly retrieve similar customer cases and recommended operational actions.

Build a semantic retrieval system that allows restaurant managers to enter natural-language queries and retrieve the most relevant historical reviews and restaurant cases along with complaint analysis, prioritization, and recommended corrective actions.

Requirement 1 (Basic)

• Basic RAG for restaurant review and complaint similarity search

• Hybrid search (keywords + semantic)

• Customer complaint classification agent

• Metadata filtering (restaurant, category, location, rating)

• Basic operational improvement recommendation

• Input validation guardrails

• Complaint prioritization

Expose the core functionality through API endpoints.

Requirement 2 (Advanced)

• DeepEval for recommendation quality evaluation

• Customer satisfaction prediction

• LLM-as-judge for complaint analysis validation

• Multi-agent restaurant operations collaboration

• Complaint trend prediction

• Review aspect and root-cause analysis

• Feedback loop for continuous improvement

Build a simple front-end interface.

Deliverables

1. Architecture Diagram

• Include a high-level view of your system in JPEG or PDF.

• Show how data is processed from the provided dataset through to the user's query response.

2. Design

• Articulate system design decisions taken and the trade-offs made.

3. Full Executable Code (Microservice)

• Focus on clarity and modularity.

• Provide a README explaining:

▪ Project setup (how to install or run)

▪ Sample usage (e.g., a test query and the resulting restaurant recommendations)

4. Panel Presentation (10 minutes)

▪ Demo of your working solution outlining the design and approach (8 minutes).

▪ Q&A (2 minutes).

Dataset Name

Yelp Open Dataset

Primary Link

Yelp Open Dataset / Yelp Data Resources

https://www.kaggle.com/datasets/yelp-dataset/yelp-dataset

Alternative Link

Yelp Review Dataset

Yelp Dataset Documentation

Format

JSON

Key Fields

business_id, name, categories, city, state, stars, review_count, attributes, hours, review_id, user_id, text, useful, funny, cool, date

Reason

Historical restaurant business information and customer reviews containing review text, ratings, categories, locations, business attributes, review dates, and customer feedback signals useful for semantic complaint retrieval, hybrid search, complaint classification, metadata filtering, prioritization, sentiment and aspect analysis, customer satisfaction prediction, complaint trend analysis, root-cause analysis, and operational improvement recommendations.